

# **EVALUATING FINANCIAL INCENTIVES IN INDIA'S DAIRY SECTOR**



**2025**

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# EXECUTIVE SUMMARY

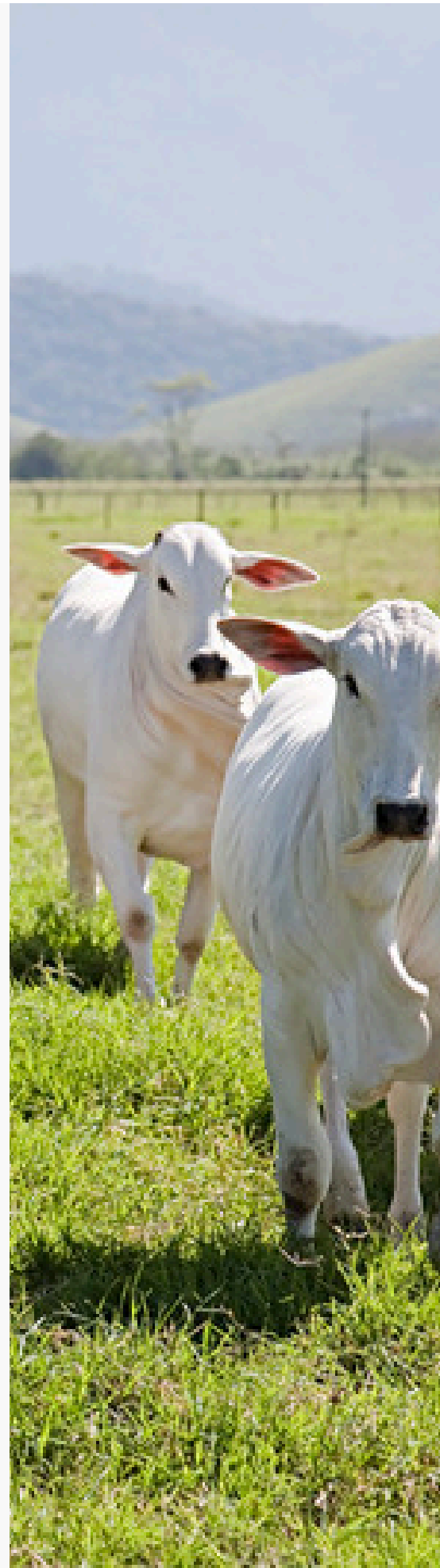
This report assesses the alignment of financial incentives within India's dairy sector with the principles of animal welfare, environmental sustainability, and public health. While extensive financial support is leading to the sector's growth, the analysis carried out demonstrates a concerning tendency for incentives to prioritize economic gains at the expense of sustainable practices. Consequently, the report highlights the persistence of issues such as poor animal welfare, environmental degradation, and potential risks to public health and food safety.

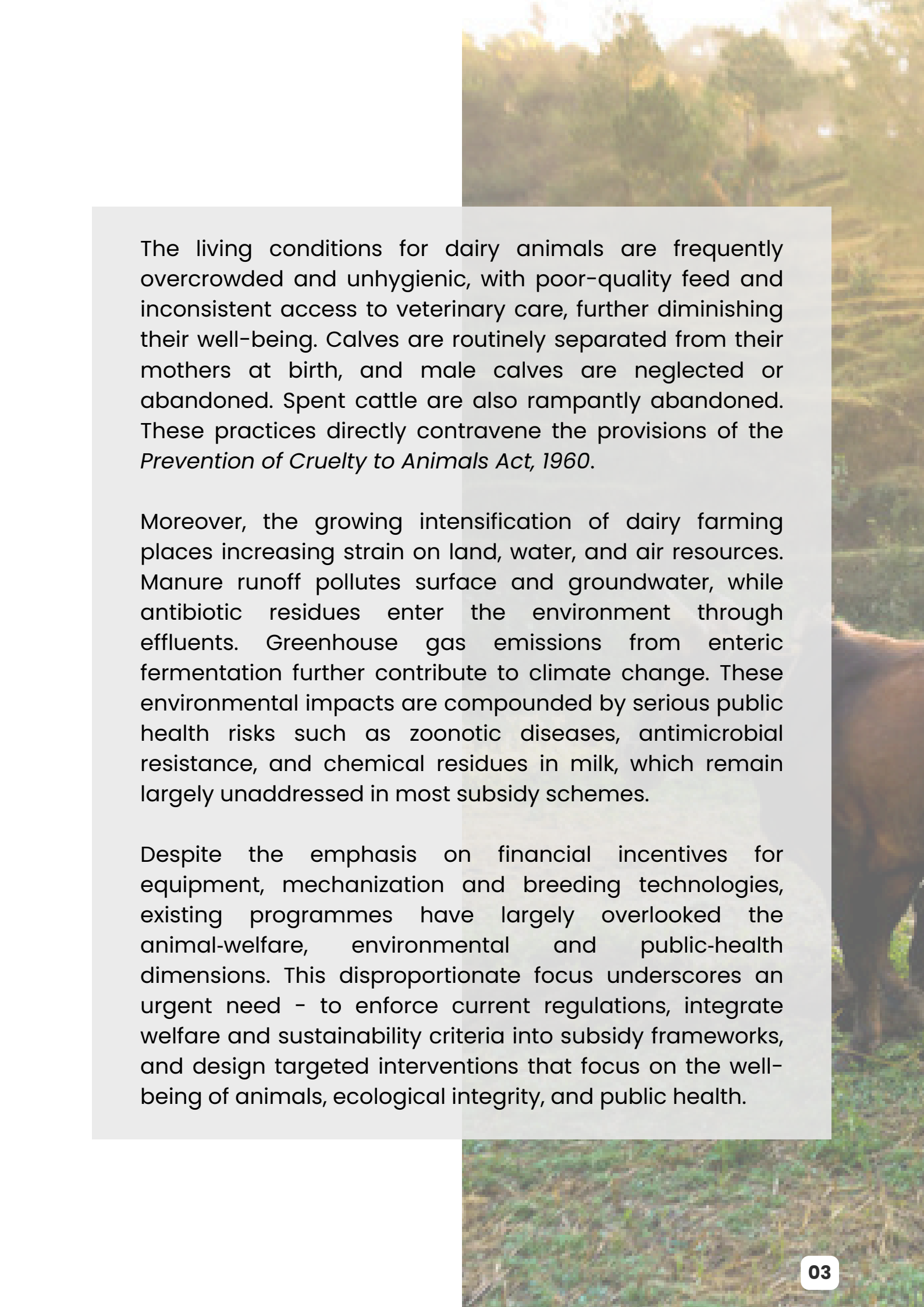
The report recommends restructuring of financial incentives to leverage them as drivers for positive change. It proposes that access to funding be linked to verifiable improvements in key areas, including animal husbandry practices, effective pollution control, and the adoption of sustainable farming methods. To achieve this transformation, the report calls for strengthened accountability and monitoring mechanisms within the financial support system, ensuring that the pursuit of economic growth in the dairy sector is effectively balanced with responsibility towards animal well-being, environmental preservation, and public health.

# INTRODUCTION

India's dairy sector is one of the largest in the world, with over 302 million bovines (cows and buffaloes) recorded in the 20th Livestock Census (2019) by the Ministry of Fisheries, Animal Husbandry and Dairying, Union of India. The Central Government has allocated substantial public funds for supporting dairy operations - approximately ₹2,910 crore between 2019 and 2024 under flagship programmes such as the *National Dairy Development Programme*, *Rashtriya Gokul Mission* and the *National Livestock Mission*. As of May 2024, banks, National Bank for Agriculture and Rural Development (NABARD) and National Dairy Development Board (NDDB) have sanctioned 408 dairy projects totaling ₹13.86 crore to bolster infrastructure, mechanization and breeding services. Additionally, the State Government offers an array of financial incentive programs to support the growth of dairy operations.

Despite rapid growth and technical advancements, on-farm realities continue to reveal widespread malpractices in animal welfare. Dairy animals are routinely subjected to excessive breeding, repeated artificial insemination, and the use of non-therapeutic hormones such as oxytocin - all of which cause significant pain and distress.





The living conditions for dairy animals are frequently overcrowded and unhygienic, with poor-quality feed and inconsistent access to veterinary care, further diminishing their well-being. Calves are routinely separated from their mothers at birth, and male calves are neglected or abandoned. Spent cattle are also rampantly abandoned. These practices directly contravene the provisions of the *Prevention of Cruelty to Animals Act, 1960*.

Moreover, the growing intensification of dairy farming places increasing strain on land, water, and air resources. Manure runoff pollutes surface and groundwater, while antibiotic residues enter the environment through effluents. Greenhouse gas emissions from enteric fermentation further contribute to climate change. These environmental impacts are compounded by serious public health risks such as zoonotic diseases, antimicrobial resistance, and chemical residues in milk, which remain largely unaddressed in most subsidy schemes.

Despite the emphasis on financial incentives for equipment, mechanization and breeding technologies, existing programmes have largely overlooked the animal-welfare, environmental and public-health dimensions. This disproportionate focus underscores an urgent need – to enforce current regulations, integrate welfare and sustainability criteria into subsidy frameworks, and design targeted interventions that focus on the well-being of animals, ecological integrity, and public health.

# OBJECTIVE

Regular review and reform of policies related to cattle welfare in India have the potential to benefit tens of millions of animals by identifying and addressing the root causes of poor welfare. Strengthening welfare standards not only improves the quality of life for animals but also contributes to better health outcomes, reduced suffering, and lower risk of zoonotic and foodborne diseases. Over time, these improvements can also help build public trust in the safety and sustainability of dairy production, encouraging more responsible consumption practices.<sup>1</sup>

The current government schemes and other financial incentives that provide extensive support to the dairy sector can be broadly classified into:

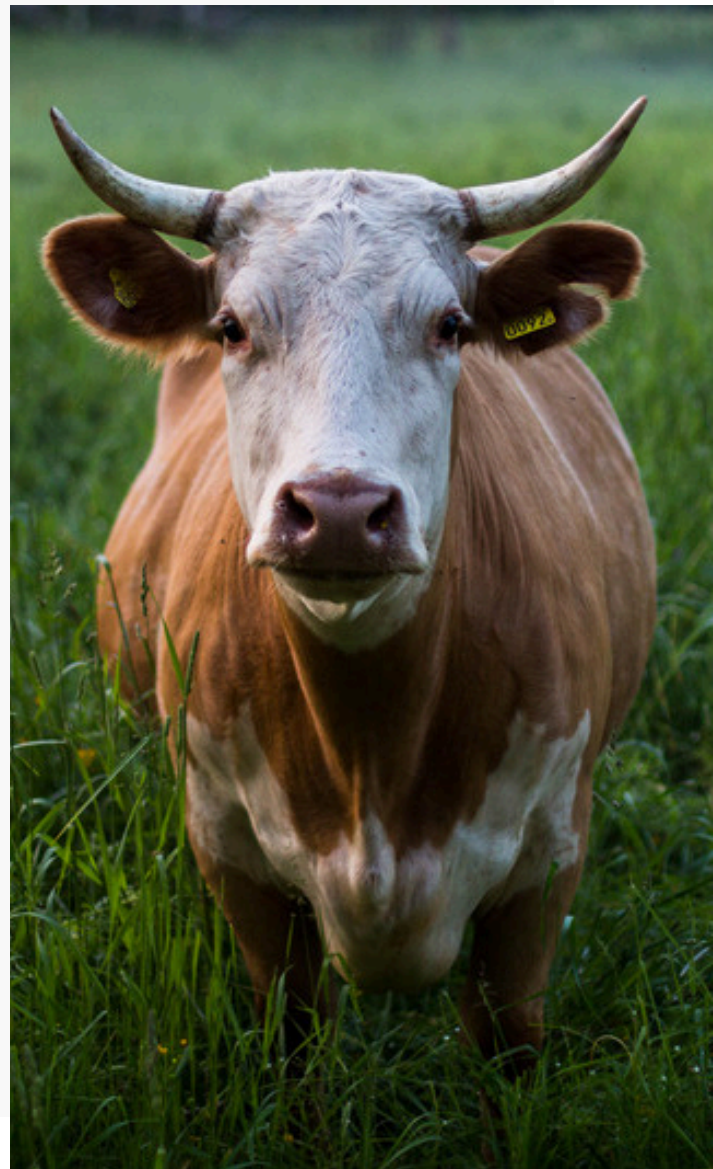
- 1. Infrastructure Development:** Financial aid for dairy processing plants, cold chain infrastructure, and equipment modernization.
- 2. Technological Advancement:** Grants for research in genetics, artificial insemination techniques, and breeding programs.
- 3. Rural Development Schemes:** Programs like the Rashtriya Gokul Mission and National Dairy Plan to enhance milk productivity and owner incomes.
- 4. Financial Inclusion:** Low-interest loans and incentives for smallholder owners under schemes like Kisan Credit Card (KCC)

# METHODOLOGY

The main focus of this report is to critically examine the financial support extended to the dairy industry – both through government schemes and subsidies, and through loans provided by banks and other financial institutions. This examination is being undertaken from the standpoint of animal protection, environmental sustainability and public health. Therefore, the scoping report required the collection of data through research of various websites of Banks, State Governments and their Departments, the Central Government and its Ministries. Furthermore, statutory laws, rules and regulations, guidelines were also sources of information for the report.

## KEY SOURCES

- Department of Animal Husbandry and Dairying
- National Dairy Development Board
- National Bank for Agriculture and Rural Development
- International financial institution
- Food and Agriculture Organization of the United Nations
- International Fund for Agricultural Development
- Animal Welfare Board of India
- State and District-Level Cooperative Milk Unions



# OVERVIEW OF REGULATORY FRAMEWORK

The regulatory framework governing dairy operations in India plays a critical role in ensuring animal welfare and promoting environmental sustainability. Aspects like housing, vaccinations, animal husbandry practices, and management of waste material, are components of financial incentives extended and are governed through many statutory provisions. These aspects can be broadly divided into two major categories, i.e., *Animal Welfare and Protection*; & *Environment and Public Health*



## ANIMAL WELFARE AND PROTECTION

### **1. The Prevention of Cruelty to Animals Act, 1960**

At the core of this framework is The Prevention of Cruelty to Animals Act, of 1960 (“PCA Act”), and the Rules formulated thereunder which set the foundation for the humane treatment and care of animals. The specific sections of the said Act are as follows:



### Section 11(1)(b)

Prohibits the employment of cattle who are old, infirm, injured, or have any disease.



### Sections 11(1)(c) & (l)

Prohibits the usage of injurious drugs or substances on cattle, including oxytocin.



### Section 11(1)(d)

Prohibits inhumane or improper transport of cattle.



### Section 11(1)(e)

Prohibits unscientific, inhumane housing conditions of cattle.



### Section 11(1)(f)

Prohibits unreasonable restriction of movement of cattle for extended periods or tethering with short ropes or heavy chains.



### Section 11(1)(h)

Prohibits negligence of cattle's food, water, and shelter needs.



### Sections 11(1)(i) & (j)

Prohibits abandonment of non-productive, sick, or injured cattle and male calves.



### Section 11(1)(k)

Prohibits sale of cattle suffering pain due to mutilation, starvation, thirst, overcrowding, or other ill-treatment by the owner.



### Section 12

Prohibits the practice of phooka or doom dev on cattle and the administration of spurious drugs, such as oxytocin.

## 2. Prevention of Cruelty to Animals (Registration of Cattle Premises) Rules, 1978

- Regulate the operation of dairies by registering premises where 5 or more cows, buffaloes, and other cattle are kept.
- The registered premises must adhere to specific parameters with regard to proper housing, feeding, and care to prevent diseases and maintain animal welfare.
- Requires a dairy unit owner/operator to prominently display Section 12 of the PCA Act in a locally understood language on the premises.

# ENVIRONMENT AND PUBLIC HEALTH

## 1. Key Environmental Acts

Under the provisions of the *Water (Prevention & Control of Pollution) Act, 1974*, and the *Air (Prevention & Control of Pollution) Act, 1981*, all dairy units will have to obtain the following permissions from the State Pollution Control Board/Pollution Control Committees

- **Consent to Establish (“CTE”)**: This consent is required to be obtained before establishing a dairy unit. The Consent to Establish is the primary clearance.
- **Consent to Operate (“CTO”)**: Once the unit is established according to mandatory pollution control systems, the units are required to obtain consent to operate.

## 2. Central Pollution Control Board’s Guidelines

The *Guidelines for Environmental Management of Dairy Farms and Gaushalas, July 2021* issued by the Central Pollution Control Board specifically address the pollution arising from dairy units.

- **Solid Waste and Waste Water Management** - Dung should be collected at regular intervals to prevent the development of obnoxious smells. The facility must be equipped to ensure proper collection, handling, and treatment of solid and liquid waste, in accordance with the scale of the dairy operations. Additionally, the flooring should be constructed from materials that prevent the percolation of wastewater into the groundwater, while also being non-slippery to ensure the safety of the animals.
- **Air Quality Management** - The facility should have adequate ventilation to prevent the build-up of greenhouse gases. Space requirements must be provided in accordance with the standards prescribed by the Bureau of Indian Standards (BIS) (IS 12237:1987).

## Space Requirements for Housing of Dairy Animals

[As per Bureau of Indian Standards (IS 12237:1987)]

| Type of Animal | Covered Area (m <sup>2</sup> /animal) | Open Paddock (m <sup>2</sup> /animal) | Maximum No. of Animals Per Pen |
|----------------|---------------------------------------|---------------------------------------|--------------------------------|
| Bulls          | 12.0                                  | 25.0                                  | 1                              |
| Cows           | 3.5                                   | 7.0                                   | 50                             |
| Buffaloes      | 4.0                                   | 8.0                                   | 50                             |
| Calves         | 1.0                                   | 2.0                                   | 30                             |
| Young Stock    | 2.0                                   | 4.0                                   | 30                             |

- **Siting Policy** - The establishment should be located at least 200 meters away from residential dwellings and 500 meters away from hospitals and schools. It must not be situated in flood-prone areas that are subject to flooding at a frequency of 1-in-25 years or more, to prevent contamination of water bodies. Locations with shallow groundwater depths of around 10 to 12 feet, especially in alluvium areas, should be avoided to prevent groundwater contamination. The site should also maintain a minimum distance of 200 meters from National Highways and 100 meters from State Highways. Additionally, it must be at least 500 meters away from major drinking water reservoirs on the catchment side, 100 meters from drinking water sources such as wells, summer storage tanks, and other tanks, 500 meters from major watercourses like rivers and lakes, and 200 meters from canals. An inter-se distance of at least 5 meters should be maintained between two establishments to allow for adequate ventilation. Each unit must also provide a setback of at least 2.5 meters on each side and develop a green belt around the facility.

## FOOD SAFETY REGULATIONS

- **Licensing and Registration** - Dairies are required to obtain either a license or registration in accordance with Section 31 of the Food Safety and Standards Act, 2006 ("FSSA Act"), based on the scale of operations.
- **Sanitary Requirements** - Ensure hygienic handling and protection of raw materials and unpacked dairy products throughout transport, loading, unloading, and storage. This includes utilizing bulk milk cooling units and employing watertight, non-corrodible containers to prevent contamination. A hygienic wastewater disposal system is essential, along with dedicated facilities for cleaning and disinfecting transport tanks. All equipment should be cleaned and disinfected after each use.
- **Personal Hygiene Requirements** - Maintain medical fitness certificates for all employees.
- **Storage Provisions** - The facility must provide clean storage for milk in steel or plastic containers, equipped with temperature control systems to keep the milk ideally at 4°C or lower. Facilities should also be in place to monitor and record temperatures, ensuring that proper storage conditions are consistently upheld.
- **Wrapping and Packaging** - The wrapping and packaging of dairy products should be carried out in hygienic rooms that are free from vermin and dust. These rooms must be equipped with appropriate tables or workstations to prevent products from being placed on the floor. The facility must also include temperature control systems and appropriate sealing devices to ensure product quality and safety.

## MUNICIPAL/LOCAL BODY STATUTES

State/municipal statutes generally empower local bodies to inspect milch cattle and regulate the infrastructure (aspects such as construction, dimensions), ventilation, lighting, cleansing-drainage, and water supply of dairies and cattle sheds. The local bodies are also authorized to regulate dairies, cattle sheds, milk shops, and stores, and to issue licenses for dairy operations, depending on the concerned area. Additionally, local bodies have the mandate to establish cattle pounds, levy penalties for allowing cattle to stray on streets or trespass on private or public property, and impose fines or seize animals found abandoned, roaming, or tethered in public spaces. They are also responsible for preventing the tethering of animals and the milking of cattle on public streets. Accordingly, every dairy owner is required to obtain a license or no-objection certificate (NOC) from the local body.



# FOCUS ON CATTLE

In India, most dairy cattle are subjected to poor welfare, despite there being established legislations and rules. Practices like artificial insemination and demands for excessive milk production lead to reduced lifespans, poor health, and suffering for dairy cattle. Unproductive animals are often neglected in overcrowded shelters or abandoned, limiting their natural behaviors. Male calves are frequently abandoned due to their low economic value. Government focus on increasing dairy production overlooks these critical welfare issues.

This report highlights the scale of suffering by analyzing dairy cattle lifespans. While naturally living up to 20 years,<sup>2</sup> intensive dairy operations drastically shorten their lives to just a few productive years due to immense strain from maximizing milk yield, leading to disease and premature culling. Some cattle are removed from production after only 2-3 years, raising serious welfare concerns and ultimately impacting productivity.

Beyond animal welfare, the dairy industry significantly contributes to environmental pollution. It generates vast amounts of wastewater, depleting oxygen in water bodies and harming aquatic life. Dairy farming also produces greenhouse gases like methane and nitrous oxide through enteric fermentation and manure management, contributing to land degradation and biodiversity loss. Addressing these issues across approximately 302 million bovines is crucial for responsible and sustainable dairy production in India

# ANIMAL WELFARE, ENVIRONMENTAL, PUBLIC HEALTH & FOOD SAFETY PARAMETERS



## A. ANIMAL WELFARE

The existing statutory mandates for dairy operations (as in the *Overview of Regulatory Framework* section) are poorly implemented. Research has found that the welfare of dairy cattle is rarely assessed in a structured manner.<sup>3</sup> Given the large scale neglect of implementation of these mandates, the scope of this report has been limited to the following-

**1. Registration under the Cattle Premises Rules:** Most dairy owners do not obtain registration under the *The Prevention of Cruelty to Animals (Registration of Cattle Premises) Rules, 1978* (“Cattle Premises Rules, 1978”). Registration of the premises is contingent upon the authorities being satisfied with the arrangements made by the owner to ascertain that the cattle are not likely to undergo any unnecessary suffering. Therefore compliance with this law inherently implies adherence to the following:

- Record keeping with regard to floor space, flooring, ventilation, supply of food and water, disinfection, drainage, disposal of dung or unwanted matter, boundary walls.
- Compulsory display of Section 12 of the PCA Act, in a language commonly understood in the locality which highlights that it is an offence if cattle are subject to the illicit administration of drugs to increase lactation/milk let down.

**Parameter** – Registration Certificate under Cattle Premises Rules, 1978

**2. Flooring:** Proper flooring is essential for maintaining hygiene and preventing injuries and disease in dairy cattle. The Central Pollution Control Board (CPCB) guidelines mandate that floors in dairy housing should be impervious, non-slippery, and equipped with a wastewater collection system. Inadequate flooring such as damp, uneven, or abrasive surfaces can lead to respiratory illness in calves and significantly increase the risk of lameness in adult cattle. In India, lameness affects an estimated 8% to 30.5% of dairy cattle and is widely recognised as the most reliable animal-based indicator of compromised welfare.<sup>4</sup>

**Parameter** – Impervious, non-slippery flooring

**3. Spurious Oxytocin:** The widespread and indiscriminate use of spurious Oxytocin, a Schedule H1 drug, to induce milk letdown in dairy animals has led to serious health issues including mastitis, abortions, delayed puberty, dystocia, and reproductive disorders such as ovarian cysts and anestrus. Its use violates Section 12 of the PCA Act, 1960, and is a cognizable offence. As per Rule 9 of the Prevention of Cruelty to Animals (Registration of Cattle Premises) Rules, 1978, Section 12 must also be prominently displayed in the local language at cattle premises to ensure compliance and deter misuse of spurious Oxytocin.

**Parameter** – Display of Section 12 of the PCA Act

**4. Access to outdoor area:** Tethering is a widespread practice in India and other parts of Asia, restricting freedom of movement and depriving cattle of exercise and natural behaviours like walking and self-grooming. Although the cattle may not exhibit an immediate physiological stress response<sup>5</sup> the long-term impact on their welfare is significant. A study in Kerala found most cattle were tied with ropes less than one metre long, often with nasal septum piercings, preventing even basic movement like turning or standing comfortably, a situation common across India.<sup>6</sup> The Bureau of Indian Standards (IS 12237:1987), referenced in the CPCB Guidelines, specifies minimum open paddock space per animal to ensure access to outdoor areas, exercise, and movement. Adhering to these standards would reduce dependence on tethering and improve overall animal welfare.

**Parameter** – Sufficient Open Paddock space per cattle

**5. Roofing:** Proper roofing is essential for protecting dairy animals from heat, rain, and environmental stress, directly impacting their health and productivity. Inadequate roofing leads to discomfort, illness, and economic losses. Ideal roofing materials should be reflective, waterproof, low in thermal conductivity, durable, and cost-effective. Additional features like insulation, shade cloth, mud plaster and green roofs can enhance comfort and sustainability.<sup>7</sup> The Bureau of Indian Standards (IS 12237:1987), referenced in the CPCB Guidelines, also prescribes minimum covered area per animal, ensuring that all cattle have sufficient shelter and are not left exposed to harsh climatic conditions.

***Parameter – Adequate Roofing***

**6. Overcrowding:** Another key welfare concern in intensive cattle operations is the use of high stocking densities, which often leads to overcrowding. This is despite the fact that the Bureau of Indian Standards (IS 12237:1987), annexed to the mandatory CPCB Guidelines, clearly specifies minimum floor space requirements per animal, along with provisions for adequate water and feeding troughs. Overcrowding increases competition for resources, leading to aggression, restricted movement, limited rest, and the inability to express natural behaviours.<sup>8</sup> These conditions not only compromise welfare but also raise the risk of disease transmission within the herd.<sup>9</sup>

***Parameter – Sufficient Housing space per cattle***

**7. Abandonment of Unproductive Cattle:** According to the National Dairy Development Board's Annual Report (2022-23), India has approximately 2,31,991 dairy cooperative societies. The NDDB has experienced significant financial growth, with its revenue rising from ₹344 crores to ₹426 crore and a net profit of ₹50 crores.<sup>10</sup> However, it is important to note that the costs associated with housing unproductive cattle are not accounted for in the budgets of the cattle owners, co-operative societies, government departments, schemes, boards or any legal entity related to the dairy industry. The following highlights the lack of accounting for unproductive cattle in the sector's overall planning and development:

- **Cruelty Towards Male Calves:** Female calves are predominantly reared as future replacement milking stock and are provided housing. However, male calves which do not produce milk for dairy operations are discarded as living by products of the dairy operations.<sup>11</sup> The cost of housing and attending to the needs of these calves is either externalized and placed upon the government/public or entirely eliminated. In both of these scenarios, the calves are subjected to practices that severely undermine their welfare.
- **Cruelty Towards Aged cattle:** The primary reasons for the abandonment of cattle in India are their unproductivity, and the maintenance costs associated with these non-productive animals. Upon abandonment, these animals roam the streets, scavenging for food from garbage and leftovers. Despite the presence of over 2,000 gaushalas<sup>12</sup> across the country, India continues to face a significant issue of stray cattle, with more than 5 million animals reported to be roaming the streets.<sup>13</sup>

**Parameter** – Space to house aged cattle and male calves

**8. Feed Storage:** It is a common practice to feed moldy domestic waste, fodder, and improperly stored feed to the lead to the cattle. However, Section 11(1) (f) of the PCA Act, requires cattle owners to cattle nutritious food. Additionally, the CPCB Guideline No.3.3 (iv) requires the feed given to be of good quality to reduce the enteric methane generation and to improve the quality of life of the cattle. Poor storage leads to mycotoxin contamination, especially Aflatoxin from *Aspergillus flavus*, which harms cattle and contaminates milk with carcinogenic toxins, posing human health risks. A study in Greater Bengaluru found widespread use of lake fodder contaminated with toxic heavy metals, highlighting the need for feed quality monitoring and secure storage to protect animal welfare and food safety to protect cattle health and ensure sustainable dairy production.<sup>14</sup>

**Parameter – Provision for secure storage of feed**

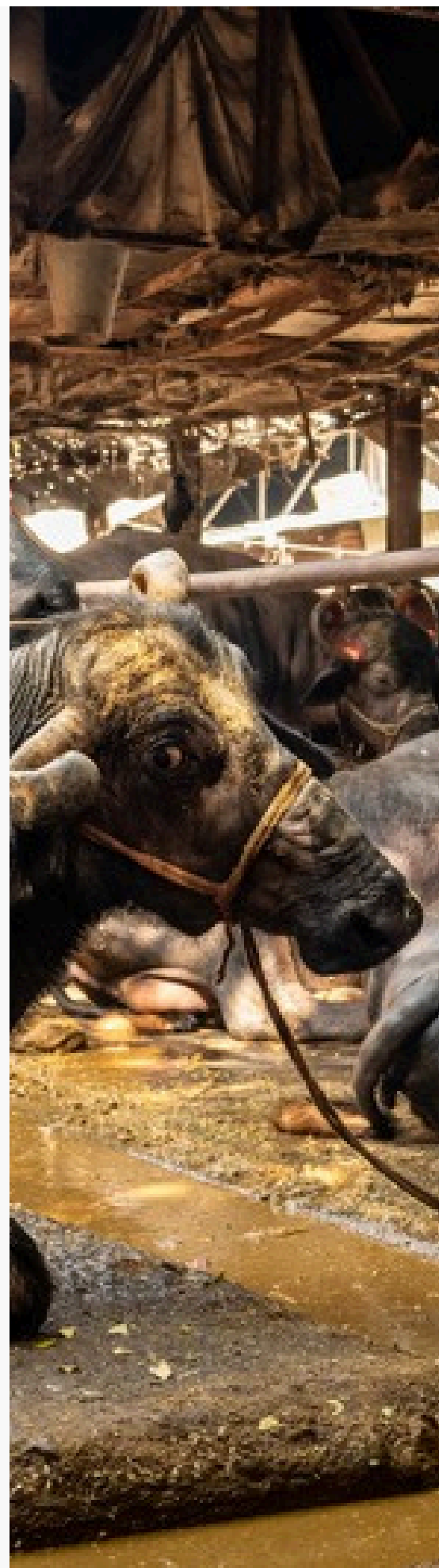
**9. Water Source:** Water is essential for lactating cows, directly impacting milk production, health, and behavior. Water intake in dairy animals is influenced by various factors such as body weight, genetic makeup, dry matter intake, feed mineral content, environmental conditions, and water temperature. It is recommended that an *ad libitum* water supply should be provided to maintain optimal health and productivity.<sup>15</sup> Moreover, Section 11(1) (f) of the PCA Act, requires cattle owners to provide with sufficient water to all the cattle. However, many dairies fail to provide sufficient water due to issues like inadequate cow-to-trough ratios and competition among cattle. In a study of 151 dairy farmers in Greater Bengaluru, only 20% of the dairy owners admitted that they were able to provide sufficient drinking water to their cattle on a daily basis.<sup>14</sup>

**Parameter – Documentation/Permission for Water Source**

## B. ENVIRONMENT

The dairy industry is one of the most polluting sectors, not only due to the volume of effluent produced but also because of its composition. Dairy processing effluents are produced intermittently, and their flow rates can vary significantly. The volume, concentration, and composition of these effluents depend on factors such as the type of product being processed, production schedules, operational methods, plant design, water and management practices.

**1. Wastewater Management:**<sup>16</sup> CPCB Guideline 5.2.2 (ii) and (iii) require dairy operators to treat wastewater in compliance with standards set by the respective State Pollution Control Boards (SPCBs) and Pollution Control Committees (PCCs). They must also prevent wastewater from seeping into the ground and contaminating groundwater. To support this, cattle shed flooring should be impervious, non-slippery, and equipped with an efficient wastewater collection system. However, most dairy operations discharge wastewater into common drains or stormwater systems, posing significant environmental risks. The following points highlight the rationale behind existing environmental regulations on dairy wastewater treatment:



## Dairy Wastewater: Key Characteristics

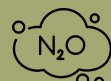
- **Effluent Parameters:** Temperature, color, pH (6.5–8.0), BOD, COD, dissolved/suspended solids, chlorides, sulfates, oil & grease – all determine toxicity.
- High in Biodegradable Organics
- Influenced by Scale & Type of Dairy Production
- **Contains Milk Constituents:** Casein & inorganic salts
- **Includes Cleaning Agents:** Detergents & sanitizers

**Parameter** – Existence of Wastewater Treatment Plant(s)

**2. Greenhouse Gas (GHG) Emissions:** Dairy farms are significant sources of anthropogenic GHG emissions. CPCB Guideline 3.3 mandates proper ventilation in animal housing to ensure fresh air circulation, reducing humidity and preventing the buildup of methane, carbon dioxide, and ammonia. Farms must also take measures to restrict pest access to stored dung to minimize odor. Improving feed quality and supplement dosage can reduce enteric methane emissions, benefiting both animal health and the environment. Additionally, planting trees or establishing green belts around dairy units helps control odors and noise pollution. These steps enhance environmental sustainability and animal welfare. The key GHG emissions from dairy farms include:<sup>17</sup>



**Methane**



**Nitrous Oxide**



**Carbon Dioxide**

**Parameter**– Adequate Ventilation Systems/Equipment

**3. Solid Waste Management:** As per CPCB Guidelines, a single bovine typically produces 15–20 kg of dung and 15–20 litres of urine daily. Guideline 3.1 requires dairy units to regularly clean shed floors, maintain hygiene in surrounding areas, and frequently sanitize and disinfect the premises. Dung must be properly collected and stored for treatment, and hazardous waste must be disposed of as per the *Solid Waste Management Rules, 2016*. Washing dung and fodder residues into drains is prohibited to prevent blockages and contamination. While CPCB recommends composting, biogas/CBG production, or dung wood manufacturing, most dairies still discharge dung with wastewater into drains, causing blockages, foul odors, and water pollution due to rapid decomposition and oxygen depletion in water bodies.<sup>18</sup>



### Impact of Solid Waste

Sewage Fungus Growth

Disease Spread

Toxicity to Aquatic Life

Oxygen Depletion

Water Quality  
Degradation

**Parameter–** Composting, Vermicomposting, Biogas, CBG production and/or Dung Wood Production

**4. Siting Criteria & Consent from SPCB/PCC:** Many dairy units in urban and semi-urban areas do not follow the siting criteria outlined in the CPCB Guidelines. Such dairies operate within residential areas, near schools and hospitals, and adjacent to stormwater drains. This not only poses significant environmental hazards but also adversely affects the welfare of the cattle. In order to ensure dairies align with siting norms, the dairy operators should obtain CTE and CTO from the SPCBs/PCCs.

***Parameters-***

4.1. CTE and CTO Certificates

4.2. 100 meters away from residential dwellings/schools/hospital

4.3. Away from the water spread area of major drinking water sources.

4.4. Away from flood plain areas of rivers and areas having shallow groundwater

4.5. At least 5 meters of inter-se distance between two establishments



## C. FOOD SAFETY AND PUBLIC HEALTH

**1. FSSAI Registration/License:** Several countries, including India, have established statutory requirements for licensing and registration of dairy operations. Depending on scale, dairy operators (barring those that supply fully to cooperative societies) need to obtain food safety licenses or registration certificates from the Central Food Safety and Standards Authority of India or the State Food Safety Department. A majority of dairy operators flout this requirement.

***Parameter-*** License/Registration from FSSAI/State Food Safety Department

**2. NOC/Permission from Local Body:** Urban Development Authorities or Municipal bodies have mandates to govern businesses/establishments that engage in food production or processing and any food-related activities. However, most dairy operators do not obtain the necessary permits to operate their units which creates public health hazards.

***Parameter -*** NOC/Permission/License from Local Body

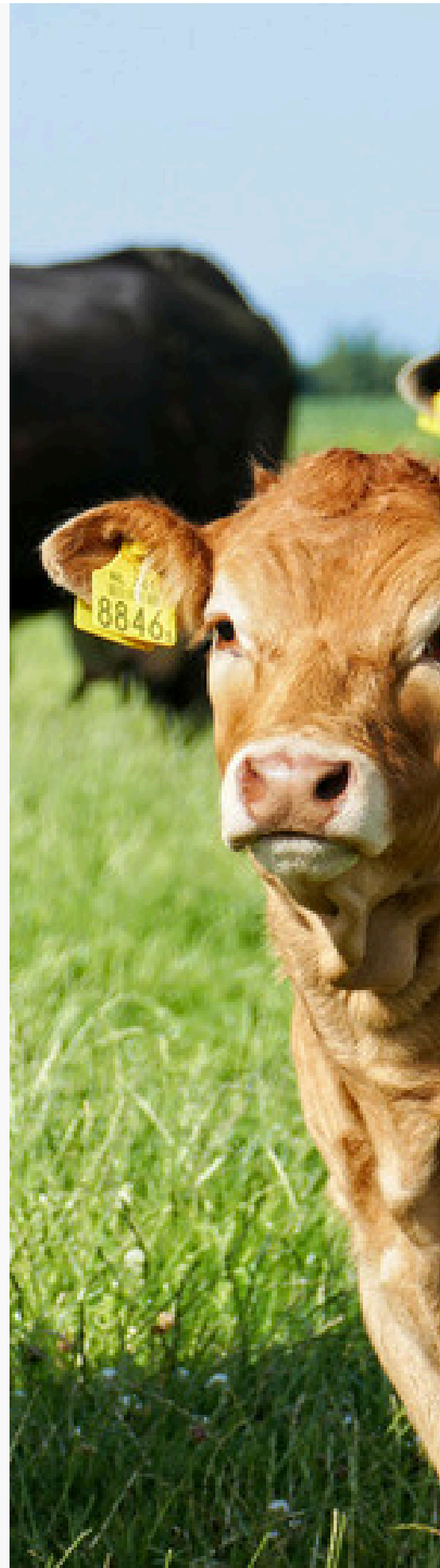


# FINANCIAL ASSISTANCE FOR DAIRY OPERATIONS

## CENTRAL GOVERNMENT SCHEMES AND SUBSIDIES

### Target Beneficiaries:

The beneficiaries for all if not most of these schemes are common and typically comprise the socially/economically marginalized groups or institutions designed to uplift such members and benefit the society at large. These include women, small land holding farmers, Farmer Producer Organizations (FPOs) and so forth.



## OBJECTIVES, COMPONENTS & SAFEGAURDS

### 1. Rashtriya Gokul Mission (RGM)

- A five-year, ₹2400 Crore (2021-2026) initiative to boost India's dairy farming.
- **Key Components:**
  - **Genetic Improvement:** Promotes high-merit germplasm, sex-sorted semen in AI, and strengthens semen stations.
  - **Breed Multiplication:** Establishment of breed multiplication farms, primarily run by state government.
  - **Artificial Insemination (AI):** Offers free doorstep services to increase coverage (30% to 70% of female cattle), including at Gaushalas (unproductive cattle sites).
  - **Animal Health & Data:** Utilizes Pashu Aadhar (INAPH) for AI-born female calf tagging, vaccinations, and deworming.
  - **Support & Policy:** Provides assistance to Gaushalas and guides policy via the Rashtriya Kamdhenu Aayog (RKA)

**Safeguards:** While the RGM entails providing large amounts of capital for undertakings such as genetic improvement, artificial insemination; there is a lack of statutory safeguards in place. The scheme does not require any clearances for animal welfare or environmental safety, barring the use of INPAH tagging. However, even INPAH tagging is limited to cover female cattle who are artificially inseminated or female calves arising from the same

## 2. National Livestock Mission

- The Budget for National Livestock Missions (“NLM”) for the year 2025-26 is Rs.800 cr which is aimed at enhancing livestock productivity across various farmed animals, including poultry, goats, pigs, and sheep.<sup>19</sup> Its key components with regard to cattle include:
  - **Overall Objective:** Increasing per-cattle productivity, enhancing fodder and feed availability, encouraging the establishment of fodder processing units and promoting risk management measures, including cattle insurance for farmers.
  - **Incentives:** 50% Capital Subsidy for infrastructure related to fodder value addition and procurement of machinery like bailers, block-making machines, TMR equipment, forage harvesters, and chaff cutters.
  - **Cattle Insurance:** Covers both indigenous and crossbred cattle (up to 5 animals per household). Requires veterinary examination, value assessment, and identification tagging with a 12-digit unique animal ID (INAPH/Pashu Aadhaar) generated by NDDB.

**Safeguards:** While the NLM entails providing large amounts of public funds for undertakings such as capital subsidy for fodder development, availability of quality feed, improvement of feed technology and cattle insurance; there is a lack of statutory safeguards barring surface level aspects. Despite significant funds for fodder and feed production, the crucial aspect of secure storage is overlooked. Without proper scientific storage to prevent infestation and mold, large investments in feed quality are futile, impacting livestock health and productivity.

### **3. National Programme for Dairy Development**

The National Programme for Dairy Development (NPDD), restructured in July 2021 (2021-26), has a total outlay of ₹1,568.28 crore, including ₹924.56 crore from a JICA ODA loan. It is not a direct beneficiary program.

- **Component A (Dairy Infrastructure Development):** Funds milk testing, chilling facilities, and farmer training for good practices, animal nutrition, and calf rearing.
- **Component B (Dairying through Cooperatives):** Enhances market access, upgrades processing, strengthens cooperatives, and trains farmers to increase milk sales.
- **Industry Mechanization Focus:** Supports installation of Automatic Milk Collection Units (AMCU), Data Processor-based Milk Collection Units (DPMCU), Bulk Milk Coolers (BMC), and related accessories/equipment.

**Safeguards:** Based on the "Operational Guidelines For Implementation Of National Programme For Dairy Development (NPDD)"<sup>20</sup> and responses to Rajya Sabha Unstarred Questions (No. 259 and 2322) by the Ministry of Fisheries, Animal Husbandry and Dairying, it appears that NPDD's July 2021 restructuring removed crucial farm-level components. In specific, the following parameters were considered in the erstwhile Operational Guidelines of NPDD but have now been omitted - 1) Composting, Vermicomposting, Biogas, CBG production, Dung wood production; 2) Adequate Ventilation Systems/Equipment; 3) Documentation/Permission for Water Source; 4) Provision for secure storage of feed; 5) Adequate Roofing, 6) Sufficient Open Paddock space per Cattle. Given the objective of the scheme has changed, the erstwhile safeguards have no applicability anymore. Thus at present, no safeguards exist in the scheme eligibility.

#### **4. Support to State Dairy Cooperative Societies and Farmers Producers Organisations**

The Support to State Dairy Cooperative Societies and Farmers Producers Organisations Scheme ("SDC & FPO") is a part of the Infrastructure Development Fund with a total allocation of ₹500 crore (2021-22 to 2025-26).

- **Objectives:** Supports dairy cooperatives and FPOs with financial relief during market distress or calamities. Enables them to procure milk at remunerative prices, especially in flush seasons.
- **Scheme Components: "Component B":** Provides Interest Subvention on Working Capital Loans for eligible dairy cooperatives and FPOs.

**Safeguards:** The only safeguard that applies for this scheme is licensing from FSSAI. However, this scheme does not require any clearances in this regard either.

#### **5. Kisan Credit Card Scheme**

The Kisan Credit Card (KCC) scheme was introduced in 1998 but was extended to dairy farmers in 2019. This extension was made to help dairy farmers meet their working capital needs.

- **Credit Facility:** Provides working capital up to ₹3 lakh (collateral-free) for dairy farmers' expenses (feed, vet care, insurance, etc.), facilitated via milk unions.
- **Tripartite Agreement:** Enables direct KCC repayment through agreements between farmers, banks, and milk unions.
- **Eligibility:** Requires milch animal details, adequate shed space proof (per NABARD guidelines), and an undertaking to replace sold animal

**Safeguards:** The scheme does not require any clearances for animal welfare or environmental safety.

## **6. Livestock Health and Disease Control Programme**

The Livestock Health and Disease Control (LH&DC) scheme and the National Animal Disease Control Programme (NADCP) were merged into the LH&DC Programme. While NADCP had a ₹13,343 Cr budget (2019 launch) and LH&DC had a ₹1470 Cr (BE)/₹886 Cr (RE) budget, operational guidelines for the merged scheme are unavailable, necessitating a two-category division for this section.

### **LH&DC Objectives & Components:**

- **Objectives:** Nationwide vaccination (FMD, Brucellosis, >100 Cr FMD doses by 2025); strengthen veterinary infrastructure; mobile vet services; improve livestock productivity/trade.
- **Key Components:** Disease control/vaccination (INAPH, campaigns); Mobile Veterinary Units (MVUs) for doorstep care; Disease Surveillance & Diagnostic Support (lab strengthening, ring vaccination).

### **NADCP Objectives & Components:**

- **Objectives:** Improve disease monitoring (serosurveillance, cold-chain, diagnostics).
- **Key Components:** Mass vaccination & health monitoring (pre-vaccination, deworming); Mandatory INAPH registration/ear tagging for vaccinated animals (100% compliance); Disease Surveillance & Diagnostic Strengthening (FMD/Brucellosis serosurveillance, ELISA labs, cold-chain); Implementation & Monitoring (district/block oversight, digital tracking).

**Safeguards:** Beyond INAPH tagging, the scheme lacks statutory animal welfare or environmental safety clearances. All cattle under this scheme require INAPH tagging for FMD vaccination.

## **7. Animal Husbandry Infrastructure Development Fund (AHIDF)**

A ₹29,110.25 crore fund extended till 2026, aimed at increasing milk and meat processing capacity by boosting investments in animal husbandry infrastructure and market access for rural producers. It merged with the Dairy Infrastructure Development Fund (February 2024). So far, 131 dairy processing projects (₹5976 Cr) and 77 breed improvement projects (₹1027.82 Cr) have been sanctioned.

- **Objectives:** Enhance farmer earnings, food security, and access to quality dairy products.
- **Key Components:**
  - **Dairy Processing Units:** Funding for new setups and upgrades, including packaging.
  - **Value-Added Dairy Products:** Support for manufacturing units (e.g., ice cream, equipment).
  - **IVF Centers:** Funding for facilities, equipment, training, and mobile labs for superior genetics.
  - **Sex-Sorted Semen Production:** Support for labs, bull sheds, research, and AI training.
  - **Breed Multiplication Farms:** Funding for infrastructure, equipment, breeding animals, and technology labs.
  - **Veterinary Vaccines & Drugs Manufacturing:** Funds facilities, cold storage, R&D, and GMP compliance.
  - **Animal Waste to Wealth Management:** Funds infrastructure for PROM, Bio-CNG, and dung/urine utilization (collection, processing, marketing).

**Safeguards:** The Scheme requires FSSAI Licensing/Registration, CTE and CTO, and Municipal License. No other safeguards are required to be adhered to.

## **8. Accredited Agent for Health and Extension of Livestock Production ("A-HELP")**

The A-HELP Programme is a joint initiative of DAHD and NRLM to generate farmer awareness. It leverages women community members, 'Pashu Sakhis,' who provide 24/7 doorstep extension services to small and marginal farmers, including dairy farmers, with priority given to Pashu Sakhis in the program's selection.<sup>21</sup>

- **Objectives:** Aims to enhance doorstep veterinary services via trained community agents, providing dairy farmers with health services, AI, disease prevention, feeding support, and insurance.

- **Key Components:**

- **Livestock Health Extension Services:** 24/7 doorstep support for vaccination, AI, INAPH tagging, disease surveillance, and basic treatment.
- **Data Collection & Digitalization:** Livestock record keeping and coordination for ear/INAPH tagging.
- **Financial & Insurance Support:** Assisting farmers with loan applications, government schemes, and insurance.
- **Sustainable Dairy Farming:** Promotes fodder cultivation and Ration Balancing Programmes for optimal nutrition.

**Safeguards:** The scheme does not require any clearances for animal welfare or environmental safety, barring support for INPAH tagging.

## BANK LOANS

Many Indian banks provide low-interest, low-margin loans to dairy owners. This section examines popular bank schemes to ascertain whether statutory safeguards are integrated into their loan sanctioning process for borrower eligibility.

### **1. Finance to Dairy Sector by Union Bank of India (UBI).**

- **Purpose:** To finance the purchase of milch cattle, machinery, utensils, shed construction, and working capital for fodder, veterinary care, and milk transportation. For small dairy units, maximum 10 animals.
- **Beneficiaries:**
  - Individual farmers (including SHGs/JLG), dairy owners (financed against milk union receivables), Corporates, FPOs/Companies, Partnership firms, and farmer Cooperatives directly engaged in agriculture/allied activities. This includes all existing and prospective dairy operators.
- **Loan Details:**

| Amount           | Loan Margin |
|------------------|-------------|
| Up to 2,00,000/- | Nil         |
| Above 2,00,000/- | Minimum 10% |

**Safegaurds:** No safegaurds to avail this loan.



## **2. Scheme for Financing Mini Dairy Units by Bank of Baroda (BOB)**

- **Purpose:** To establish new small dairy units with 2 to 10 milch animals
- **Beneficiaries:** Individuals, Farmers, Members of NGOs/SHGs/JLGs.
- **Document Required:** KYC Documents (AADHAR, Voter ID, PAN Card, Driving License), Passport Size photo, Land Records and Quotation/Invoice (if available)
- **Loan Details:**

| Amount           | Loan Margin |
|------------------|-------------|
| Rs 3,00,000/-    | Nil         |
| Above 3,00,000/- | Minimum 10% |

**Safeguards:** No safeguards are required to avail this loan.

## **3. Dairy Loans by Canara Bank (CB)**

- **Purpose:** To finance milch cattle purchase, dairy machines, shed construction, on-farm processing, and green fodder cultivation. Also covers working capital (feed, vet, milk transport) and animal transportation costs.
- **Eligibility:** Individual farmers (including SHGs/JLG), NGOs, companies, and organized/unorganized groups (e.g., dairy cooperatives, milk unions). Dairy owners can be financed against receivables from milk unions/cooperatives.

| Amount           | Loan Margin      |
|------------------|------------------|
| Up to 2,00,000/- | Nil              |
| Above 2,00,000/- | Minimum 15 - 25% |

**Safegaurds:** While the loan specifies basic requirements like cattle sheds, fodder arrangements, and water availability, these are vague and do not constitute effective safeguards. Consequently, this bank requires no true safeguards for loan sanction.

#### 4. HDFC

- **Purpose:** A term loan is given to dairy farmers for purchase of milch cattle basis recommendation/guarantee from dairy societies or private dairy companies.
- **Eligibility:** Based on credit history from milk payments/receipts. Individual dairy farmers aged 18-60 years, owning at least 2 cattle with an average annual milk collection of ₹80,000 in the last year.
- **Farmer Documentation:** Required documentation includes KYC for all parties, a 1-year milk payment history, land ownership records, and passport-size photographs.
- **Loans:**
  - **Funding Dairy Societies (HDFC 1):** The Cash Credit facility given to Dairy Co-operative societies to meet their working capital requirements.
  - **Dairy Plus Term Loans (HDFC 2):** These are Term Loans extended to the dairy societies for forward lending to its farmer members for purchase of new processing machines and advancement of the technology etc.
  - **Dairy Power (HDFC 3):** It is aimed to provide the necessary liquidity to the farmers to sustain in the cattle rearing.

**Safeguard:** The loan requires a minimum land holding of 0.5 acre required in the name of applicant or family member. However, this does not comprise effective safeguards because it does not specify the number of cattle that can be housed in the given area. Therefore, this bank effectively does not require any safeguards to be in place to avail the loan.

## 5. State Bank of India (SBI)

### A. KCC for Animal Husbandry & Fisheries

- **Purpose:** Working capital for milch animal rearing.
- **Eligibility:** Farmers (individual/joint), JLGs, SHGs, including tenant farmers with sheds.
- **Documents:** Basic KYC, photos, address/ID/dairy activity proof.

### B. Pradhan Mantri Mudra Yojana (PMMY)-Allied Agri

- **Purpose:** Employment and income generation to improve livelihoods.
- **Eligibility:** Farmers (individual/joint), JLGs, SHGs, including tenant farmers with sheds.
- **Documents:** Application, ID proof (Voter ID/PAN/Aadhaar/Driving License).
- **Loan Details:**

| Amount                              | Loan Margin      |
|-------------------------------------|------------------|
| Up to 50,000/-                      | Nil              |
| Above 50,000/- and Up to 1,00,000/- | Minimum 15 - 25% |

### C. Agri & Food Enterprises Loan (AFEL)

- **Purpose:** Finance for agro-based enterprises (commodity, food processing, agri-input, exporters, FPOs, infrastructure).
- **Eligibility:** Proprietorships, Partnerships, Pvt Ltd, LLPs, Corporates, FPOs, Cooperatives.
- **Documents:** SME Series (standard business documents).
- **Loan Details:**

| Amount                                       | Loan Margin                                               |
|----------------------------------------------|-----------------------------------------------------------|
| Upto ₹5 Cr if Credit guarantee cover availed | Nil                                                       |
| More than ₹5 Cr                              | Suitable security for rest of the loan amount above ₹5 Cr |

**Safegaurds:** No safegaurds for any of the loans.

## **6. Bank of India: Doodhganga Scheme**

- **Purpose:** Finances milch animal purchase, new/expanded dairy units, calf rearing, crossbreeding, milk machinery, shed construction/renovation, and all dairy equipment/utensils.
- **Eligibility:** Individuals, SHG/JLG groups, Cooperative societies, Companies, Associations, partnership firms, proprietorship concerns/FPOs/FPCs.
- **Documents Required:** KYC (ID/address), proof of land holding, and evidence of adequate know-how, experience/training, and infrastructure.
- **Loan Details:** No margin requirement for loans up to 2,00,000/-

**Safeguards:** No safeguards are required to avail this loan.

## **7. Agriculture Finance Dairy Farming – IDBI Bank Dairy Loans**

- **Purpose:** Credit for individuals and groups to purchase high-yielding milch animals (indigenous and exotic breeds), construct sheds, acquire dairy equipment (e.g., chaff cutters), and cover animal transportation costs.
- **Eligibility:** Individuals and groups of farmers experienced and actively engaged in dairy farming.
- **Bank Loan Details:**

| Extent of Exposure  |
|---------------------|
| Minimum 20,000/-    |
| Maximum 10,00,000/- |

**Safegaurds:** Although the loan finances construction of cattle sheds, it cannot be considered a safeguard because the space requirements as per the BIS standard is not being prescribed and does not comprise effective safeguards because they are vague in nature. Therefore, this bank does not require any safeguards to be in place to avail the loan.

## **8. Scheme For Financing Dairy Development Programmes by Punjab National Bank (PNB)**

Comprehensive financing for dairy development, encompassing milk production, calf rearing, cattle breeding, processing, and allied activities.

- **Key Financing Areas:**

- **Milk Production:** Purchase and maintenance of high-yielding milch animals (various breeds), shed construction, and dairy machinery/equipment.
- **Calf Rearing:** Support for rearing good quality female calves up to first lactation.
- **Processing & Infrastructure:** Funding for milk-processing facilities and construction of milk houses.
- **Other Activities:** Includes cattle breeding (AI), salvaging dry pregnant/male calves, and pasture development.

- **Eligibility:**

- **General:** Experienced individuals/groups engaged in dairying (main/subsidiary), with feed/fodder storage arrangements.
- **Specific (Calf Rearing):** Cattle breeders/progressive farmers committed to retaining calves for milk or selling at first calving, with land for fodder. Projects require techno-economic viability.

**Safeguard:** Although the scheme finances construction of cattle sheds, it cannot be considered a safeguard because the space requirements as per the BIS standard is not being prescribed and therefore, does not comprise effective safeguards because they are vague in nature. Therefore, this bank does not require any safeguards to be in place to avail the loan.

## **9. DAIRY – (Purchase of Milch Animals) : Indian Overseas Bank**

- **Objective:** Credit for establishing/expanding dairy farms: financing milch animals, sheds, water facilities, dairy equipment, fodder farms, and female calf rearing (Heifer Calf Rearing Scheme). Also considers financing dairy cooperative milk houses.
- **Eligibility:** Applicants should have aptitude and experience in dairy farming/milch cattle rearing, sufficient space for animals, and regular water and fodder supplies.
- **Loan Details:**

| Amount           | Loan Margin      |
|------------------|------------------|
| Up to 2,00,000/- | Nil              |
| Above 2,00,000/- | Minimum 15 – 25% |

**Safegaurds:** Despite financing sheds and water, the loan lacks effective safeguards due to vague requirements, such as unspecified BIS space standards and unclear definitions of 'adequate' water facilities. Therefore, it does not comprise effective safeguards because they are vague in nature

## **10. Dairy Loan – Karur Vysya Bank**

Credit for individuals and groups to purchase high-yielding milch animals (indigenous and exotic breeds), construct cattle sheds, and acquire dairy equipment (e.g., chaff cutters), including animal transportation costs.

- **Eligibility:** Individuals, farmers, groups of farmers, firms, companies, societies, NGOs.
- **Loan Details:**

| Limit          | ROI%                  |
|----------------|-----------------------|
| Up to 50,000/- | One year MCLR + 0.90% |
| Above 50,000/- | One year MCLR + 0.90% |

**Safegaurds:** No safeguards are required to avail this loan.

## 11. Central Bank

### A. Cent Dairy Scheme

- **Purpose:** Establishment of dairy units for milk production.
- **Eligibility:** Individuals, farmers, groups of farmers, firms, companies, societies, NGOs etc.
- **Loan Details:**

| Amount                           | Loan Margin |
|----------------------------------|-------------|
| Up to Rs 1,00,000/-              | Nil         |
| 1,00,000/- and Up to 5,00,000/-  | 10%         |
| 5,00,000/- and Up to 10,00,000/- | 15%         |
| Above 10,00,000/-                | 20%         |

**Safeguards:** No safeguards are required to avail this loan.

### B. Cent Animal Husbandry Infra Scheme

- **Purpose:** Dairy processing and value addition infrastructure
- **Eligibility:** Individual entrepreneurs, private companies, MSME, Farmer Producer Organizations and Section 8 companies

| Entity                      | Loan Margin |
|-----------------------------|-------------|
| Micro and small enterprises | 10%         |
| Medium enterprise           | 15%         |
| Others                      | 20%         |

**Safeguards:** No safeguards are required to avail this loan.

# ASSESSING FINANCIAL INCENTIVES FOR WELFARE & SUSTAINABILITY

The table evaluates financial incentives related to animal welfare, sustainability, and statutory parameters, analyzing their impact on animal welfare, the environment, and food safety.

| Parameter | Safeguard                                                                 |
|-----------|---------------------------------------------------------------------------|
| P1        | Registration Certificate under Cattle Premises Rules, 1978                |
| P2        | Impervious, non-slippery flooring                                         |
| P3        | Display of Section 12 of the PCA Act                                      |
| P4        | Open Paddock for Each Cattle                                              |
| P5        | Adequate Roofing                                                          |
| P6        | Sufficient housing minimum space per cattle                               |
| P7        | Sufficient Open Paddock space per Cattle                                  |
| P8        | Adequate Roofing                                                          |
| P9        | Space to house aged cattle and male calves                                |
| P10       | Provision for secure storage of feed                                      |
| P11       | Documentation/Permission for Water Source                                 |
| P12       | Existence of Wastewater Treatment Plant(s)                                |
| P13       | Adequate Ventilation Systems/Equipment                                    |
| P14       | Composting, Vermicomposting, Biogas, CBG production, Dung wood production |
| P15       | Watertight, non-corrodible containers, waste water disposal system        |
| P17       | CTE, CTO                                                                  |
| P18       | FSSAI License/Registration                                                |
| P19       | Municipal License                                                         |

## CENTRAL GOVERNMENT SCHEMES

| Parameter | RGM | NLM | NPCC | SDC & FPO | LHDCP | AHIDF | KCC | A-HELP |
|-----------|-----|-----|------|-----------|-------|-------|-----|--------|
| P1        | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P2        | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P3        | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P4        | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P5        | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P6        | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P7        | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P8        | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P9        | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P10       | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P11       | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P12       | ×   | ×   | ×    | NA        | NA    | ✓     | ×   | NA     |
| P13       | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P14       | ×   | ×   | ×    | NA        | NA    | ✓     | ×   | NA     |
| P15       | ×   | ×   | ×    | NA        | NA    | ×     | ×   | NA     |
| P17       | ×   | ×   | ×    | NA        | NA    | ✓     | ×   | NA     |
| P18       | ×   | ×   | ×    | NA        | NA    | ✓     | ×   | NA     |
| P19       | ×   | ×   | ×    | NA        | NA    | ✓     | ×   | NA     |

**\*NA** - Not applicable because the given parameter does not directly pertain to the components under consideration of the particular Scheme.

## BANK LOANS

| P   | UBI | BOB | Can B | HDFC 1 | HDFC 2 | HDFC 3 | SBI1 | SBI 2 | SBI 3 | BOI | PNB 1 |
|-----|-----|-----|-------|--------|--------|--------|------|-------|-------|-----|-------|
| P1  | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P2  | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P3  | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P4  | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P5  | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P6  | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P7  | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P8  | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P9  | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P10 | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P11 | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P12 | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P13 | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P14 | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P15 | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P17 | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P18 | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |
| P19 | ×   | ×   | ×     | ×      | ×      | ×      | ×    | ×     | ×     | ×   | ×     |

## BANK LOANS

| P   | PNB 2 | PNB 3 | PNB 4 | IDBI | IOB | KYB | Cent B 1 | Cent B 2 |
|-----|-------|-------|-------|------|-----|-----|----------|----------|
| P1  | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P2  | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P3  | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P4  | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P5  | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P6  | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P7  | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P8  | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P9  | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P10 | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P11 | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P12 | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P13 | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P14 | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P15 | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P17 | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P18 | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |
| P19 | ×     | ×     | ×     | ×    | ×   | ×   | ×        | ×        |

# CONCLUSION

Of the 8 Schemes examined, only 1 scheme - AHIDF Scheme covered any of the parameters under consideration. With regard to the 19 bank loans studied, it was noted that not a single bank had incorporated the key parameters identified as safeguards in their loan processes.

Analysis of existing schemes and bank loan provisions has revealed key insights into the current framework's limitations in promoting cattle welfare within the dairy sector. The absence of mandatory animal welfare and sustainability criteria is not just an ethical concern but a pressing national issue with far-reaching consequences such as:

**1. Environmental Impact:** Non-compliant dairy units pose a significant threat to the environment. Extending loans at concessional rates or offering subsidies without adequate safeguards will inevitably support practices that are environmentally harmful.

**2. Public Health:** Unregulated dairy farming often leads to unsanitary conditions, heightening the risk of zoonotic diseases, antibiotic resistance, directly endangering public health. The illness of cattle due to aflatoxin exposure increases the risk of cancer in humans who consume such contaminated milk.<sup>22</sup>

**3. Animal Welfare:** Many non-compliant dairy units routinely engage in practices that disregard basic animal welfare and violate existing legal protections. Without accountability, these operations continue to inflict avoidable suffering on animals. In contrast, responsible dairying is the only path to ensuring that milk production remains both environmentally and ethically sustainable for future generations.

**4. Failure to Meet Global Commitments:** As a signatory to the United Nations Sustainable Development Goals (SDGs), India must ensure that its agricultural policies align with targets such as clean water (Goal 6), responsible production (Goal 12), and climate action (Goal 13). The absence of sustainability measures in financial support schemes hinders our progress toward these global benchmarks.

|                                                                                                           |                                                                                                        |                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
|                        |                     |                                                  |
| Preventing water contamination from dairy waste safeguards local water bodies and drinking water sources. | Encouraging ethical and sustainable livestock farming ensures resource efficiency and waste reduction. | Mitigating the environmental impact of dairy farming helps reduce greenhouse gas emissions and supports long-term climate resilience. |

To address these urgent challenges, financial aid for dairy operations must be contingent upon adherence to necessary animal welfare, food safety and environmental standards. Without integrating existing legal documentation requirements into financial support schemes, India risks reinforcing outdated and harmful farming models rather than fostering a future of responsible and sustainable agriculture.

# RECOMMENDATIONS

This section highlights these gaps and proposes actionable recommendations to address them, all of which can be implemented within the scope of the existing statutory framework.



## **1. Strengthening Legal and Regulatory Compliance:**

- Making it mandatory for all dairy operators to submit documents pertaining to the four statutory requirements (from the local body, food safety authority, registering authority for cattle premises and pollution board board/committee) to avail benefits should be incorporated.
- All Central and State Schemes should incorporate the 19 key welfare parameters detailed in this report.
- Reserve Bank of India (RBI) should issue directions to all banks across the country to ensure safeguards described in this report are incorporated in loan requirements.
- Department for Promotion of Industry and Internal Trade (DPIIT) should introduce animal welfare due diligence as a criterion for approving Foreign Direct Investment proposals in dairy farming.

## **2. Responsible dairying:**

- The Polluter Pays Principle (PPP), endorsed by the OECD in 1972, requires polluters to bear the cost of managing harm caused by their operations. In India's dairy sector, this includes soil degradation, excessive water use, pollution, GHG emissions, and the neglect of male calves and unproductive cattle. Commercial dairy enterprises routinely externalize these costs, especially by abandoning non-milking animals.
- Financial support for dairying should be contingent on the dairy operators undertaking to develop infrastructure and/or arrangements for rearing male calves and unproductive female cattle. It should also be undertaken that feed, healthcare, and shelter for these cattle shall be ensured. This will promote ethical dairy practices, prevent cattle abandonment, and reduce public health burden.

## **3. Capacity Building, Training and Awareness:**

- Capacity building/training of financial institutions and their personnel to ensure that aid extended follows from ensuring humane animal husbandry practices, disease prevention, promotion of scientific waste management and methane management by educating dairy stakeholders on emissions, water use, soil pollution, and efficient resource use should be incorporated.

## **4. Monitoring and Enforcement:**

- Implement a system of monitoring and enforcement to ensure compliance with animal welfare and sustainability standards by agents extending financial aid, through regular inspections and third-party audits of dairy operators.
- Making it a precondition for incentives extended, for dairy operators to ensure that their cattle are identified with INAPH tags.

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We are prepared to assist in structuring financial incentives to ensure sustainable growth and positive impact on animal welfare and the environment within India's dairy sector.



Plot No. 26, D.D.A. Opp Gulmohar Enclave Gate  
No. 3, New Delhi - 110049



+91 88262 89111; +91 99999 66304



[corporate.communications@varahaimpact.com](mailto:corporate.communications@varahaimpact.com)